

Principal Component Regression (PCR)

Understanding When Dimensionality Reduction Works — and When It Fails

TTK4260 — Assignment

Scenario: You are a data scientist at *NordicEnergy Analytics*, working on renewable energy forecasting for the Norwegian power grid. Your task is to predict hourly output from wind turbines (**WindPower**) and solar panels (**PVPower**) using weather station measurements. The predictor matrix $\mathbf{X}$ contains 14 weather variables including radiation, wind speed, temperature, and atmospheric conditions. These predictors are highly correlated, making ordinary least squares unstable. You will investigate whether Principal Component Regression (PCR) can reduce dimensionality while maintaining predictive accuracy — and discover a fundamental limitation of this approach.

Question 1 (Part 1: Setup, Data Loading, and Initial Exploration)

The dataset consists of hourly measurements from a Norwegian weather station paired with energy production data. The predictor matrix $\mathbf{X}$ contains 14 weather variables, and the response matrix $\mathbf{Y}$ contains two targets: **WindPower** and **PVPower**.

- (a) Import the required libraries (`numpy`, `pandas`, `matplotlib`, `seaborn`, `sklearn`). Load the data from `./data/X.txt` and `./data/Y.txt`. Show the shape of $\mathbf{X}$ and list all 14 feature names. [4 points]
- (b) The variable `IsDayBin` is categorical (“Day” or “Night”). Explain how you are going to deal with this variable. [3 points]
- (c) Standardize the predictor matrix using `StandardScaler` to obtain $\mathbf{X} \in \mathbb{R}^{n \times 14}$ with mean 0 and standard deviation 1 for each column. Verify numerically that standardization was successful. [4 points]
- (d) Compute and visualize the correlation matrix of $\mathbf{X}$ as a heatmap. Identify and report three pairs of variables that show strong positive correlation ($r > 0.8$). Explain what this multicollinearity implies for ordinary least squares regression. [5 points]

Question 2 (Part 2: Multiple Linear Regression (MLR) Baseline)

Before applying PCR, establish a baseline using ordinary least squares. Let $\mathbf{y}_{\text{wind}}$ and $\mathbf{y}_{\text{pv}}$ denote the two response vectors.

- (a) Write down the ordinary least squares solution for regressing $\mathbf{y}$ on $\mathbf{X}$:

$$\hat{\boldsymbol{\theta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

Explain why the invertibility of $\mathbf{X}^\top \mathbf{X}$ may be problematic when predictors are highly correlated. [5 points]

- (b) Fit two separate linear regression models: one for **WindPower** and one for **PVPower**. Report the coefficient of determination (R^2) for both models. [4 points]
- (c) Implement a bootstrap stability test: resample the data 200 times, fit MLR on each bootstrap sample, and collect the coefficients. Create side-by-side boxplots of the bootstrapped coefficients for both responses. [6 points]

- (d) Compute the condition number of $\mathbf{X}^\top \mathbf{X}$ using `np.linalg.cond()`. A high condition number indicates multicollinearity. Interpret your result and relate it to the coefficient stability observed in the boxplots. [5 points]

Question 3 (Part 3: PCA on X — Understanding the Variance Structure)

Apply PCA to the standardized predictor matrix $\mathbf{X}$ to understand its variance structure.

- (a) Write down the PCA decomposition and state the dimensions:

$$\mathbf{X} = \mathbf{TP}^\top + \mathbf{E},$$

where $\mathbf{T} \in \mathbb{R}^{n \times r}$ are the scores and $\mathbf{P} \in \mathbb{R}^{14 \times r}$ are the loadings. Explain what the first principal component $\mathbf{t}_1$ represents in terms of variance maximization. [6 points]

- (b) Fit a full PCA model (`PCA()` with no constraint on the number of components). Extract the eigenvalues λ_i and compute the explained variance ratio for each component. Report these values for the first 6 components in a table. [5 points]
- (c) Create a scree plot showing eigenvalues vs component number. Identify a potential “elbow” and propose a candidate number of components r based on this criterion. [4 points]
- (d) Create a cumulative explained variance plot. Determine the smallest r such that cumulative explained variance exceeds 60%. Compare this to your choice from the scree plot and justify a final selection of $r = 3$ components. [5 points]

Question 4 (Part 4: Principal Component Regression with $r = 3$)

Now build PCR models using the first $r = 3$ principal components as predictors.

- (a) Describe the PCR procedure: (1) fit PCA on $\mathbf{X}$ to get scores $\mathbf{T}$, (2) regress $\mathbf{y}$ on the first r score columns, (3) transform coefficients back to the original feature space. Write the prediction equation:

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}_{\text{PCR}}, \quad \hat{\boldsymbol{\theta}}_{\text{PCR}} = \mathbf{P}_r \hat{\boldsymbol{\theta}},$$

where $\hat{\boldsymbol{\theta}}$ are the regression coefficients in PC-space. [6 points]

- (b) Implement PCR with $r = 3$ for both responses. Report the R^2 values for `WindPower` and `PVPower`. [4 points]
- (c) Create a comparison table showing MLR vs PCR($r = 3$) performance:

Method	R^2 (WindPower)	R^2 (PVPower)
MLR (14 features)	...	...
PCR ($r = 3$ components)	...	...

Comment on any surprising asymmetry in performance between the two responses. [5 points]

- (d) Create predicted vs actual scatter plots for both responses under PCR($r = 3$). Add a diagonal reference line ($y = \hat{y}$) and visually assess the quality of fit. Explain the dramatic difference in prediction quality. [5 points]

Question 5 (Part 5: PCA Diagnostics — Score Plots)

Use PCA score plots to diagnose why PCR works well for one response but fails for the other.

- Create a scatter plot of the first two PC scores (t_1 vs t_2) with points colored by **WindPower** magnitude. Use a continuous color scale (e.g., **viridis**). Describe any visible gradient or structure. [4 points]
- Create a second score plot (t_1 vs t_2) colored by **PVPower** magnitude. Compare the structure to the previous plot. Which response shows a clear relationship with the first principal component? [4 points]
- Based on the score plots, explain in 3–5 sentences why $\text{PCR}(r = 3)$ is effective for **PVPower** but ineffective for **WindPower**.

Hint: Consider where the predictive information resides in PC-space. [6 points]

Question 6 (Part 6: Correlation Loading Plot — The Key Diagnostic)

The correlation loading plot reveals which original variables are captured by each PC, and where the response variables project onto the PC subspace.

- Compute the correlation between each original variable x_j and each PC score t_i :

$$\rho_{j,i} = \text{Corr}(x_j, t_i) = \frac{\text{Cov}(x_j, t_i)}{\sigma_{x_j} \sigma_{t_i}}.$$

Since both are standardized, this simplifies to the dot product. Also compute the correlation of each response with the PC scores. Store results in a matrix. [5 points]

- Create a correlation loading plot for PC1 vs PC2:

- Plot a unit circle
- Plot arrows from the origin to each variable's coordinate
- Color-code: radiation variables in orange, wind-related in blue, others in gray
- Overlay the two responses as star markers
- Add appropriate labels and legend

[7 points]

- Identify which cluster of variables lies near **PVPower** on the plot. Explain why this means PC1 captures the solar signal well. [4 points]
- Identify where **WindPower** lies on the plot. Explain why being near the origin indicates that PC1 and PC2 do not capture the wind signal. [4 points]

Question 7 (Part 7: Feature–Response Relationships)

Investigate the direct relationships between individual features and the two responses to understand the underlying physics.

- Create scatter plots showing:

- **WindPower** vs **WindSpeed**
- **WindPower** vs **IrrDirect**
- **PVPower** vs **IrrDirect**
- **PVPower** vs **WindSpeed**

Use a 2×2 subplot layout. Add trend lines using linear regression. [6 points]

- (b) Based on these plots, explain which predictor(s) drive **WindPower** and which drive **PVPower**. Relate this to the physical nature of wind turbines and solar panels. [5 points]
- (c) Compute and report the Pearson correlation between:
- **WindPower** and **WindSpeed**
 - **PVPower** and the four radiation variables

Explain why **PVPower** benefits from a *correlated block* of predictors, while **WindPower** depends on a single predictor. [5 points]

Question 8 (Part 8: What Do PC1 and PC2 Actually Capture?)

Examine the loading vectors to interpret what each PC represents in terms of the original variables.

- (a) Create loading bar charts for PC1 and PC2 showing the contribution of each original variable. Use horizontal bars sorted by magnitude. [5 points]
- (b) Interpret PC1: Which variables have the largest (in magnitude) loadings? What physical phenomenon does PC1 represent?
- Hint: Look at the radiation variables.* [4 points]
- (c) Interpret PC2: Which variables dominate PC2? Explain how PC2 differs from PC1 in terms of what it captures. [4 points]
- (d) Compute the correlation between each response and all 14 PC scores. Create a bar plot showing $|\text{Corr}(y, t_i)|$ for each response. Identify which PC ranks have the strongest correlation with **WindPower**. Explain why these high-correlation PCs are not among the first three retained in $\text{PCR}(r = 3)$. [7 points]

Question 9 (Part 9: PCR Accuracy vs Number of Components)

Investigate how prediction accuracy changes as more PCs are added to the model.

- (a) Implement a loop over $r = 1, 2, \dots, 14$. For each r :
- Fit PCR with the first r components
 - Compute R^2 for both **WindPower** and **PVPower**

Store the results. [5 points]

- (b) Plot R^2 vs number of components for both responses on the same axes. Use different colors and markers for the two responses. [4 points]
- (c) Identify the minimum number of components needed to achieve $R^2 > 0.80$ for **WindPower**. Compare this to the number needed for **PVPower**. [4 points]
- (d) Explain why **WindPower** requires more components than **PVPower** to achieve good predictions. Relate this to the location of the wind signal in PC-space (refer to your answer from Question 8d). [5 points]

Question 10 (Part 10: Conclusion and the Fundamental Lesson)

Synthesize your findings and articulate the central limitation of PCR.

- (a) Write down the objective function that PCA maximizes:

$$\max_{\|\mathbf{p}_a\|=1} \text{Var}(\mathbf{X}\mathbf{p}_a) = \max_{\|\mathbf{p}_a\|=1} \mathbf{p}_a^\top \mathbf{S} \mathbf{p}_a,$$

where $\mathbf{S}$ is the covariance matrix of $\mathbf{X}$. Explain what this means: PCA seeks directions of maximum variance in $\mathbf{X}$. *[5 points]*

- (b) For supervised learning (regression), what should we actually maximize to find good predictive directions?

$$\max_{\|\mathbf{p}_a\|=1} \text{Cov}(\mathbf{X}\mathbf{p}_a, \mathbf{y}) = \max_{\|\mathbf{p}_a\|=1} \mathbf{p}_a^\top \mathbf{X}^\top \mathbf{y}.$$

Explain why these two objectives are not the same. *[5 points]*

- (c) Summarize the key findings from this investigation in a short paragraph (5–8 sentences):

- What worked and what failed in PCR($r = 3$)?
- Why did it fail for one response but not the other?
- What is the relationship between X-variance and predictive power?

[6 points]

- (d) State the fundamental limitation of PCR in 1–2 sentences. Mention that this motivates a different method (Partial Least Squares, PLS) that directly maximizes covariance between $\mathbf{X}$ and $\mathbf{y}$. *[4 points]*